

Lab 3b – Introduction to Objects Input and Output

Working with Input and Output Files

Question 1 : Employee Payroll

Input: A file named **employees.txt** containing an employee's first name, last name, hourly rate, and hours worked.

Example input:

John Smith 25.50 40

Mary Jones 30.00 35

David Brown 22.75 42

Your name and hourly rate and hours worked

Task: Employee Gross Pay Calculation

- a. Write an algorithm to solve this problem
- b. Write a Java program that:
 1. Uses FileReader and Scanner to read the data from employees.txt.
 2. Calculates each employee's gross pay:
gross pay = hourly rate × hours worked
 3. Uses PrintWriter to write the results to **payroll.txt**.
 4. Uses printf to format the output so that:
 - o Employee name is displayed in a fixed-width field.
 - o Hourly rate and gross pay are displayed with 2 decimal places.
 - o Hours worked are displayed with 1 decimal place.
 5. Closes both files.

Expected output format:

Employee Rate Hours Gross Pay

John Smith 25.50 40.0 1020.00

Mary Jones 30.00 35.0 1050.00

David Brown 22.75 42.0 955.50

Your name and your proposed pay rate and gross pay

Question 2: Student Grade Report

Input: A file named **grades.txt** containing a student's first name, last name, and five test scores.

Example:

Michael Johnson 85 92 78 88 95

Sarah Williams 90 84 91 87 93

Yourname with proposed 5 test scores

Provide 2 additional names of your choice and 5 test scores

Task: Develop an algorithm and then write a Java program that reads student information from **grades.txt** and five test scores from an input file, calculates the student's average score, and writes a formatted grade report to an output file gradeReport.txt.

- a. Write a step-by-step algorithm to solve this problem
- b. Write a Java program that:
 1. Uses Scanner to read the student information from **grades.txt**.
 2. Calculates the average of the five test scores.
 3. 6. Use printf() formatting to create a neatly aligned report. The column headers (Student, Test1, Test2, Test3, Test4, Test5, and Average) must be centered above their respective columns.
 4. Uses printf to produce a neatly formatted report.
 5. Displays the average to 2 decimal places.
 6. Closes both files.

Expected output (gradeReport.txt):

Student	Test1	Test2	Test3	Test4	Test5	Average
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Michael Johnson	85	92	78	88	95	87.60
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Sarah Williams	90	84	91	87	93	89.00
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Submission Requirements

Submit the following:

1. Your completed algorithms.
2. Your Java source codes (.java file).
3. Your input files.
4. Your output files.
5. Upload all required files to your assigned GitHub repository and submit the GitHub repository link.
6. Verify that the files in GitHub are the final and most recent versions of your work.